Lecture 7 (Thermo)

Tuesday, 31 January 2023 10:31 AM

Stationary states : solutions to TISE $\quad i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$

$\hat{H}\Psi = E_n \Psi_n$; $\Psi_n \rightarrow$ stationary state (physically observable properties are constant)

Basis Set

Basis functions to solve K-S eqns ; Larger basis set → more accurate solutions, but more computational time } w/ convergence tests [Consistent for all DFT Codes]

Basis set size given as energy cutoff :

PIB : $E_{pw_j} = \frac{\hbar^2 \pi^2}{2m}\left(\left(\frac{n_x}{L_x}\right)^2 + \left(\frac{n_y}{L_y}\right)^2 + \left(\frac{n_z}{L_z}\right)^2\right)$; pw → plane wave

$n_x, n_y, n_z = 1, \ldots \infty$

Choose total plane wave energy "cut off energy" → Determine no. & type of plane waves that fall in this E bound

→ In 1D, for a unit w/ d ~ 1nm, usually 4-6 basis functions for convergence → E ~ 10-15 eV } Codes can be sped up w/ 1000 of tricks

→ In 3D, unit cell V ~ $1\ nm^3$, thousands of planewaves needed ~ cutoff ≈ 400 eV

Recall K-S eqns:

$$\left[-\frac{1}{2}\nabla_i^2 + V_{ext}(r) + \int \frac{\rho(r')}{|r-r'|}dr' + \frac{\delta E_{xc}}{\delta \rho}\right]\Psi_i = \varepsilon_i \Psi_i$$

↑ This is where most approximations are made

Possible choices:

1) LDA → Local density approximation : (very expensive)

Similar to class functionals → function of ρ(r) only

Fit to QMonte Carlo calc. of homogenous e^- gas

$E_{xc}^{LDA} = E_{xc}^{LDA}[\rho]$ (Closed form)

+ Works best for bulk materials/systems w/ slowly varying (wrt space) ρ's.
+ Does not work well for surfaces or systems w/ sharp ρ gradients

2) Generalized Gradient Approximation (GGA)

All are explicit in ρ and $\nabla\rho$: $E_{xc}^{GGA} = E_{xc}^{GGA}(\rho, \nabla\rho)$

+ Standard for surfaces and interfaces
ex: PBE → Purdue-Burke Ernzerhoff
RPBE → Revised "
PW91 →
BEEF → Bayesian estimate

3) Meta-GGA

Incorporates second derivatives : $E_{meta} = E_{meta}[\rho, \nabla\rho, \nabla^2\rho]$

ex:
+ SCAN
+ Improvement in performance is relatively small

4) Hybrid/Hyper functionals

Based on GGA functionals, but add a fraction of exact exchange from Hartree Fock

Calculate HF exchange using Ψ_i from DFT

$E_{xc}^{hyb} = E_x^{GGA} + a\left(E_x^{HF} - E_x^{GGA}\right) + E_c^{GGA}$

↳ E_{xc} depends on Ψ_i's so leads to much longer computation times

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Ex: B3LYP, M06, PBE0, HSE06
↳ a=0.25 ↳ Hybrid screen exchange (a=0.25) w/ Ex applied over short distances

+ Full hybrids don't work well on metals because e^- clouds screen out exchange effects
+ Screen hybrids are better but not great
+ Hybrids work well for oxides (esp semiconductors)

Major limitations of existing DFT techniques:

1) Band gaps (HOMO-LUMO gaps) in semiconductors or molecules:

$\Delta = \varepsilon_{valence} - \varepsilon_{conduction}$

$\varepsilon_{LUMO} - \varepsilon_{HOMO}$

Typically, $\Delta_{GGA,\ LDA} < \Delta_{exp}$ because they don't localize e^- properly (i.e. errors in e^- localization)

2) Self-interaction error:

In DFT, e^- feel themselves;

Exact Coulomb: $V = \frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}\frac{e^2}{4\pi\varepsilon_0|r_i - r_j|}$

DFT: $\frac{1}{2}\iint \frac{\rho(r)\,\rho(r')}{|r - r'|}\,dr\,dr'$ (no $j\neq i$ equivalent)

Therefore, the issue is DFT under localizes e^-; for metals errors cancel.
However, in oxides, redox processes give errors that do not cancel (as much)

Most imp. for d and f e^-'s which are highly localized

Two practical approaches:

① DFT + U (Hubbard U) → empirical term that penalizes delocalized orbitals; U depends on material

② Exact exchange functionals: SIE is smaller because of Hartree-Fock (SIE exactly cancels in HF)

③ Van der Waals interactions: In standard GGA calculations $\rho(r)$ decays exponentially from nucleus; } exponential is faster
vDW decays as $\frac{1}{r^6}$

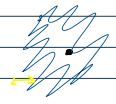
intermolecular interactions are underestimated

Empirical corrections:

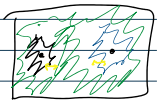
(a) DFT-Dx ; x=1,2,3

+ Add $\frac{C_i}{r^6}$ terms to energy
+ Not great for metals (does not account for e^- screening)





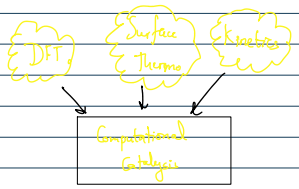
+ Correlated fluctuations in e^- density result in vdW
+ In metals, this is reduced due to the presence of metallic delocalized e^-



VdW is essential for zeolites, hydrocarbon interactions on surface.

(b) Explicit non-local functionals that include vdW like-terms (→ self-consistently incorporated into energies) → ok for metals

Eg: DFT-vdW, opt PBE, opt BBB, BEEF-vdW



Surface Thermodynamics

DFT + Thermo gives: ① What is state of cat. surface (structure, ads coverage, e^- properties) for given environmental conditions? (T, P_i, V)

② How can we determine adsorption isotherms for use in microkinetic models?

For surfaces, convinient free energy is "Grand Potential"

$$\Omega = \underbrace{E - TS}_{\text{Helmholtz energy (F)}} - \sum_i \mu_i N_i \quad ; \quad \text{FE: } d\Omega = -SdT - PdV - \sum N_i d\mu_i$$

(Ω, S, V, N are extensive)

T, V, μ_i are fundamental variables

Convinient free energy because P is hard to define for a surface

Also, surface is often in quasi-eq w/ gas or solid bulk phase → easier to find μ_i's

Consider: 

If interphase is planar:

$T^\alpha = T^\beta$; $P^\alpha = P^\beta$; $\mu_i^\alpha = \mu_i^\beta$

$$\Omega^{total} = \Omega^\alpha + \Omega^\beta + \Omega$$

↳ excess grand potential from interface